

The Gut–Brain Axis in Dogs and Cats During International Transport: Neuroendocrine Integration, Microbiota and Energy Metabolism

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Abstract

Objective. To integrate mechanistic evidence on the gut-brain axis in canids and felids subjected to international transport, with emphasis on neuroendocrine activation, microbiota changes and energy metabolism alterations. **Method.** Technical review of published literature on the HPA axis, sympathoadrenomedullary (SAM) system, stress-induced dysbiosis, tight junction permeability, secretory IgA, feline hepatic lipidosis and psychobiotics, integrating rodent models and direct studies in dogs and cats. **Results.** Transport simultaneously activates the SAM system (catecholamines) and the HPA axis (cortisol), altering intestinal motility, tight junction permeability and mucosal immunity. Stress-induced dysbiosis differs between canids and felids in composition and recovery rate; cats present a specific risk of hepatic lipidosis from prolonged fasting. Evidence on psychobiotics and prebiotics for gut-brain axis modulation in dogs and cats is preliminary and has not yet translated into validated travel protocols. **Conclusion.** The gut-brain axis constitutes an integrated stress-response system during transport; its understanding is clinically relevant for designing protocols that minimize metabolic, microbial and immunological impact in pets during international travel.

Keywords: gut-brain axis, dogs, cats, international transport, microbiota, neuroendocrine stress, HPA, dysbiosis, hepatic lipidosis, psychobiotics

Section 2 Transport as a multisystemic stressor in dogs and cats

The classification of transport as a "stressor" in veterinary medicine does not refer to an inferred subjective experience, but to the documented capacity of specific travel conditions to activate physiological stress-response systems in dogs and cats. This distinction is methodologically relevant: neuroendocrine activation can occur — and has been measured — in the absence of overt behavioural signs of distress, meaning that an animal's apparent calm during transport does not exclude activation of the SAM and HPA axes (Wormald et al., 2017).

The stressors identified in the context of international transport include components of different natures acting simultaneously, and their interaction has not been studied in an integrated fashion in either dogs or cats. Three main categories are described:

Physical stressors: mechanical vibration from the vehicle or aircraft, noise levels above those of the habitual environment, temperature and humidity variations, and relative oxygen partial pressure reductions in cargo holds (documented for brachycephalic dogs in Article III of this series). In normocephalic animals, relative cabin hypoxia has not been specifically studied as an HPA axis activator.

Psychosocial stressors: confinement to a reduced space, separation from the primary attachment figure, exposure to unfamiliar olfactory and auditory stimuli, and unpredictability of the environment. These factors overlap with components of what the literature terms "uncontrollable stress" — stress whose defining characteristic is not intensity but the animal's absence of control (Hennessy et al., 2001).

Metabolic stressors: pre-transport fasting recommended for management reasons, hyporexia during travel, altered feeding rhythms and, in prolonged journeys, potential relative dehydration. These factors act on the same neuroendocrine systems as psychosocial stressors, potentiating their effects on the gastrointestinal tract.

Journey duration introduces a critical variable: available studies in dogs show that the cortisol response peaks in the first 20–30 minutes of transport and tends to decline — though not always to baseline values — in the hours that follow (Prpar-Mihevc et al., 2017). However, in journeys lasting several hours, the persistence of physical and psychosocial stressors may sustain activation levels above baseline for extended periods; on transmeridian flights, circadian desynchronization (jet lag) adds to the picture. No studies in cats have measured the temporal dynamics of cortisol specifically during long-haul air transport.

Section 3 Neuroendocrine activation during confinement and fasting

3.1 Sympathoadrenomedullary (SAM) system

The sympathoadrenomedullary system represents the fastest response pathway to acute stress. Upon perception of a threat or a novel stimulus, the sympathetic nervous system activates the adrenal medulla, which releases epinephrine (adrenaline) and norepinephrine (noradrenaline) into the systemic circulation. This process occurs within seconds and its systemic effects are immediate: increased heart rate and cardiac output, redistribution of blood flow away from viscera toward skeletal muscle and the heart, hepatic glucose mobilisation, and inhibition of digestive processes.

In the gastrointestinal tract, sympathetic activation produces splanchnic vasoconstriction — reduction of blood flow to the intestinal mucosa — and motility inhibition. This effect is immediate, reproducible and well characterised in mammals, including studies in dogs in which epinephrine infusion reproduces the motility changes observed during acute stress (Ruckebusch & Buéno, 1976, seminal; confirmed in subsequent reviews). Dog Direct evidence in dogs for splanchnic vasoconstriction specifically during transport is indirect; it is inferred from cortisol values and the observation of gastrointestinal symptoms (vomiting, diarrhoea) in dogs during travel, without direct measurement of splanchnic flow in that context.

In cats, sympathetic activation during transport has been documented by measurement of plasma catecholamines and cardiovascular markers in a seminal study by Hydbring-Sandberg et al. (2004), which recorded immediate glucose peaks and cortisol responses upon movement onset in 12 cats. Cat Extrapolation of the specific gastrointestinal effects of that activation in cats during long-haul transport exceeds available evidence and is declared as a physiologically plausible inference, not a documented finding.

3.2 Hypothalamic-pituitary-adrenal (HPA) axis and cortisol

The HPA axis acts in parallel with the SAM system but with a slower kinetics. Under sustained or anticipatory stress, the hypothalamus releases CRH (corticotropin-releasing hormone), which stimulates the pituitary gland to secrete ACTH, which in turn activates the adrenal cortex to produce glucocorticoids — primarily cortisol in dogs and cats, with species differences in the cortisol-to-corticosterone ratio.

In dogs, cortisol elevation during transport has been directly documented. Wormald et al. (2017) measured salivary cortisol in 24 dogs during terrestrial transport and recorded statistically significant elevations ($p < 0.05$) in the first 10–20 minutes. Dog Prpar-Mihevc et al. (2017) documented that serum cortisol levels doubled after two hours of road transport in 15 non-habituated dogs, returning to baseline values at 24 hours. These studies confirm HPA axis activation in the context of terrestrial transport; extrapolation to long-haul air transport introduces additional uncertainty that has not been directly studied. Declared extrapolation

Glucocorticoids exert multiple systemic effects that are relevant to intestinal function. At the mucosal level, sustained cortisol suppresses secretory IgA production, reduces epithelial cell proliferation, inhibits mucus synthesis by goblet cells and modifies the expression of tight junction proteins — effects described in rodent models and, with more indirect evidence, in dogs hospitalised under chronic stress. Rodent Extrapolation to Canid

CRH, in addition to its action on the pituitary, acts directly on the enteric nervous system through CRH type 1 receptors (CRH-R1) present in the intestinal mucosa. This local pathway — independent of systemic cortisol — accelerates colonic transit and increases paracellular permeability, which explains the acute diarrhoea observed in stress situations without necessarily involving sustained cortisol elevation. This pathway is well documented in rodent and human models; its relevance in dogs and cats is inferred from the presence of CRH receptors in canine intestinal mucosa, though the specific functional evidence in these species is limited. Rodent Human Extrapolation to Canid

Section 4 Intestinal effects of acute stress

Gastrointestinal motility

Intestinal motility is regulated by the enteric nervous system (ENS), whose activity is modulated in real time by sympathetic, parasympathetic and humoral signals. Acute stress produces a well-documented biphasic pattern in animal models: acceleration of colonic transit (explaining stress-related diarrhoea) combined with inhibition of gastric emptying and slowed small intestinal transit. In dogs, vomiting and diarrhoea during and after transport are frequently reported clinical observations. However, studies that have directly measured motility parameters — gastric emptying time, intestinal transit speed — during transport in dogs are scarce, and available data derive primarily from confinement or noise stress models. Dog

Mucosal immunity: secretory IgA and immune cells

Secretory IgA (sIgA) is the principal antibody of mucosal surfaces and plays a fundamental role in microbiota regulation and defence against luminal pathogens. Its production depends on plasma cells in the lamina propria and on transcytosis through the intestinal epithelium — an energetically costly process that is suppressed under sustained glucocorticoid stress.

In humans, reduction of salivary and intestinal sIgA has been documented under acute and chronic psychological stress. Human In rodents, confinement stress significantly reduces intestinal sIgA levels within hours, with partial recovery upon cessation of the stressor. Rodent In dogs and cats, there is no direct evidence measuring intestinal sIgA during or after transport. The inference that prolonged transport suppresses mucosal IgA in these species rests on the mechanistic plausibility of the glucocorticoid effect, not on species-specific evidence. Extrapolation — no direct evidence in dogs/cats during transport

The gut-brain axis also involves changes in the proportion of lamina propria immune cells: acute stress mobilises neutrophils into the circulation, reduces the proportion of regulatory T lymphocytes in the intestinal mucosa, and alters the function of antigen-presenting dendritic cells. These effects have been documented in rodent models and in some studies of hospitalised dogs, but not specifically in the transport context. Rodent Dog (hospitalisation)

Intestinal permeability and tight junctions

Tight junction proteins — including occludin, claudins and members of the ZO family — form the paracellular seal of the intestinal epithelium. Their integrity is essential for the epithelium to function as a selective barrier, permitting absorption of nutrients and water while preventing the passage of macromolecules and microorganisms from the lumen into the systemic circulation.

Tight junction protein expression is reduced under sustained exposure to glucocorticoids and CRH, with consequent increase in paracellular permeability. This phenomenon is well characterised in vitro and in murine models. Rodent In dogs, some studies on chronic enteritis and hospitalisation stress have documented mucosal barrier alterations. Dog (clinical conditions) The application of this mechanism to the transport context in healthy animals is a physiologically plausible inference whose direct validation would require specific studies not currently available.

Section 5 Microbiota under transport stress

5.1 Evidence from rodent models

Rodent stress models have been the most widely used to study the bidirectional relationship between the nervous system and the intestinal microbiota — the so-called microbiota-gut-brain axis. Studies of maternal separation stress, immobilisation and chronic low-grade stress in mice and rats have consistently documented: reduced microbial diversity, decreases in short-chain fatty acid (SCFA)-producing bacteria such as *Lactobacillus* and *Bifidobacterium*, and relative increases in Gram-negative bacteria with pro-inflammatory potential. Rodent The direct relevance of these models for dogs and cats subjected to transport cannot be assumed without caution: differences in baseline microbiome composition between laboratory rodents and domestic carnivores are substantial, and the stress conditions in murine models differ qualitatively from those of transport.

5.2 Evidence in dogs

The most relevant evidence in dogs regarding microbiota changes under environmental stress comes from studies in animals in shelter and hospitalisation contexts. Pilla and Suchodolski (2020) reviewed in *Frontiers in Veterinary Science* the determinants of the canine microbiome and documented that acute stress exposure is associated with reduced microbiome richness and alterations in the Dysbiosis Index (DI) — a validated metric that evaluates the balance between bacterial taxa associated with health and those associated with dysbiosis. In particular, a decrease in butyrate-producing bacteria, including *Faecalibacterium*, was observed. Dog

Stiefelmaier et al. (2020) documented in *PLOS ONE* that environment-change stress in shelter dogs was associated with an increase in *Fusobacteria* and a reduction in alpha diversity during the first seven days of housing. Dog It is important to note that conditions in a shelter — multiple simultaneous stressors, high density, pathogen exposure — differ from the context of transport with an owner; the magnitude of changes may not be comparable.

Vázquez-Baeza et al. (2016) published data on the canine microbiome in a hospitalisation context, documenting a compositional microbiome shift under clinical environment stress even in the absence of antibiotics. Dog [DOI pending direct editorial verification before final publication]

5.3 Evidence in cats

5.4 Differences and gaps: summary

The canine microbiota is the best-studied among domestic carnivores, and a validated instrument for measuring dysbiosis exists (canine DI). Available evidence suggests — with moderate level of evidence, based on studies in contexts of environmental stress, not transport specifically — that stress is associated with compositional changes in the canine microbiome. For cats, the gap is practically total. No identified study has measured feline microbiota during or immediately after transport. The assumption that changes parallel those observed in dogs or rodents is an extrapolation that cannot be made without explicit declaration. Non-validated inter-species extrapolation

Section 6 Metabolic stress and fasting

6.1 Metabolic adaptation in dogs

Dogs display a well-documented capacity for metabolic adaptation to short- and medium-term fasting that reflects their evolutionary history as a species with irregular food access. Upon fasting, hepatic glycogen depletion occurs in the first 12–24 hours, after which hepatic gluconeogenesis from amino acids, lactate and glycerol increases. Simultaneously, fatty acid mobilisation from adipose tissue increases the availability of alternative energy substrates. Dog

In the transport context, pre-travel fasting recommended to reduce the risk of vomiting and aspiration — typically 4–12 hours depending on journey duration — combined with hyporexia during travel results in periods of caloric restriction that, in healthy adult dogs with adequate body condition, do not typically represent a serious metabolic risk. This statement is limited to healthy animals; geriatric individuals, those with pre-existing metabolic conditions, or those with reduced muscle mass may have a compromised adaptive response to fasting.

Elevated cortisol during stress additionally contributes to glucose and fatty acid mobilisation, an effect that is adaptive in the short term but that, when sustained, can contribute to muscle protein catabolism. In long-haul journeys, this catabolic contribution may be clinically relevant in animals with limited reserves, although no quantitative studies in dogs under real transport conditions have directly measured protein catabolism markers.

6.2 Feline metabolic vulnerability: the hepatic lipidosis risk

Cats differ fundamentally from dogs in their lipid metabolism. As obligate carnivores with constitutively active protein metabolism, cats have a limited capacity to reduce amino acid oxidation when protein intake decreases, generating early protein depletion during fasting. Simultaneously, fatty acid mobilisation from adipose tissue occurs more rapidly than in dogs, and the capacity of feline hepatocytes to handle a massive influx of free fatty acids is limited.

The result of this combination — rapid lipid mobilisation plus limited hepatic processing capacity — is intrahepatic triglyceride accumulation, the process that defines feline hepatic lipidosis. Biourge et al. (1994) experimentally documented that in overweight cats, detectable biochemical changes compatible with hepatic lipid accumulation can appear after approximately 48 hours of absolute fasting. The complete clinical syndrome, with jaundice and severe hepatic enzyme elevation, was precipitated by periods of total anorexia of 5 to 7 days in overweight cats. Cat

It is important to qualify this risk precisely. The 48-hour threshold for detectable biochemical changes was documented in overweight cats under conditions of controlled absolute fasting. Cats making a journey of fewer than 24 hours, with partial intake before and after, and without excess body weight, have a substantially different risk profile. Risk stratification requires consideration of: total fasting duration, body condition score, underlying health status, and post-travel feeding behaviour. This analysis exceeds the scope of this article and corresponds to individual veterinary assessment. Cat

Section 7 Evidence on psychobiotics and microbiota modulation

Psychobiotics have been defined as live microorganisms that, when administered in adequate quantities, confer a mental health benefit to the host through modulation of the microbiota-gut-brain axis (Dinan et al., 2013). This definition, originating in human medicine, has been adapted in the veterinary context to refer to probiotic strains whose primary effect of interest is behavioural or neuroendocrine rather than digestive.

State of evidence in dogs

The body of evidence on psychobiotics in dogs is nascent and methodologically heterogeneous. The study with the highest available level of evidence is McGowan (2016), a blinded controlled trial evaluating administration of *Bifidobacterium longum* (BL999) in dogs with anxiety behaviours, documenting reduction in salivary cortisol and decreases in vocalisations and anxiety behaviours. Dog — RCT, relatively high evidence level for the field

Batool et al. (2021) evaluated *Lactobacillus rhamnosus* in dogs and reported improved reactivity to unfamiliar stimuli after four weeks of administration. Bray et al. (2021) studied *Bifidobacterium animalis* without finding significant effects on cognition, with a trend toward temperament stabilisation under moderate stress. Dog

State of evidence in cats

Evidence on psychobiotics specifically in cats for stress or anxiety modulation is even more limited than in dogs. Existing studies on probiotics in cats focus primarily on digestive outcomes (diarrhoea, microbiota composition). Extrapolation of gut-brain axis effects from the dog studies to cats is not supported by species-specific evidence. Cat — evidence gap

Prebiotics: brief consideration

Prebiotics — primarily fermentable fibres such as fructooligosaccharides (FOS) and inulin — modulate microbiota composition by favouring the growth of SCFA-producing bacteria. Their effect on the gut-brain axis is indirect and mediated by the SCFAs themselves, which exert effects on the vagus nerve and on enterochromaffin serotonin production. In dogs, the effect of prebiotics on stress markers has not been directly studied. Their inclusion in this article is limited to description of the mechanism of action, without formulating a use recommendation. Mechanism described; effect on canine/feline stress not directly validated

Section 8 Mechanistic synthesis in the context of transport

International transport of dogs and cats activates, simultaneously and potentially synergistically, the SAM and HPA axes. SAM activation is immediate — in the order of seconds to minutes after the onset of confinement — and produces splanchnic vasoconstriction, motility inhibition and suppression of digestive processes. HPA axis activation, slower, sustains and amplifies these effects through circulating cortisol, with a documented peak in the first 20–30 minutes in dogs under terrestrial transport and with a temporal dynamics not yet studied in dogs under long-haul air transport nor in cats under any type of prolonged transport.

Sustained cortisol exerts effects on the intestinal barrier through two parallel pathways: direct, by suppressing tight junction protein expression and reducing epithelial proliferation; and indirect, through immune suppression that reduces available sIgA on the mucosal surface. Hypothalamic CRH additionally acts locally on enteric receptors accelerating colonic transit and increasing paracellular permeability, independently of systemic cortisol. This mechanism is well documented in rodents and humans; in dogs and cats it is inferred from the presence of CRH receptors in intestinal mucosa, without specific functional evidence during transport. Declared extrapolation

The intestinal microbiota is sensitive to stress-induced changes in the intestinal environment — modifications in luminal pH, motility, secreted antimicrobial peptides and availability of fermentable substrates — and responds with detectable compositional changes. In dogs, studies in environmental stress contexts document Dysbiosis Index alterations and reduction of butyrate producers. These changes have not been studied in the specific context of transport, and no equivalent evidence exists for cats. Recovery of the baseline microbiota composition after cessation of the stressor is variable and has not been characterised in the transport context in either species.

In cats, the specific metabolic vulnerability is hepatic triglyceride accumulation in response to hyporexia. The biochemical mechanism is well documented in fasting models: rapid fatty acid mobilisation from adipose tissue, insufficient hepatic oxidation capacity to handle the influx, and progressive intrahepatic accumulation. Post-transport stress hyporexia may be a contributing factor in susceptible individuals — especially overweight cats — though the direct causal chain transport → lipidosis has not been documented in prospective studies. Veterinary consideration of feeding behaviour in the days following travel in overweight cats or those with a history of stress-related hyporexia is a reasonable clinical inference, not a claim based on direct evidence.

Section 9 Limitations of current knowledge

The limitations of the state of science in this field are substantial and must be declared precisely for the reader to calibrate the weight of each claim in this article.

Scarcity of studies in real transport contexts. The majority of studies on stress in dogs and cats use laboratory models — confinement stress, separation, noise, hospitalisation — that differ qualitatively from international transport. The few studies that have measured cortisol during transport are limited to short-duration terrestrial journeys. No longitudinal studies have measured neuroendocrine, intestinal or microbiota markers in dogs or cats during and after a long-haul flight.

Species-level evidence asymmetry. The canine microbiome is significantly better studied than the feline. Psychobiotic studies in cats in the context of stress and transport are practically non-existent. Claims about cats in this review rest more frequently on mechanistic plausibility than on direct species-specific evidence.

Between-individual heterogeneity. The neuroendocrine response to stress varies substantially between individuals within the same species, as a function of prior experience, habituation to transport, temperament, body condition score and underlying health status. Available studies do not permit the construction of individualised risk profiles.

Limitations of microbiota studies. Intestinal microbiota characterisation using 16S rRNA sequencing detects compositional changes but does not directly inform metabolic function. Available studies do not consistently distinguish between transient changes (hours to days) and sustained microbiota alterations.

Absence of bacterial translocation studies in transport. No evidence exists of clinically relevant bacterial translocation in healthy dogs or cats during standard transport. Presenting this phenomenon as an expected consequence of travel would not be supported by available literature.

Nascent state of psychobiotic evidence. The number of controlled trials in dogs is small, sample sizes are modest, and the heterogeneity of strains, doses and outcomes evaluated prevents formulation of evidence-based recommendations.

Section 10 Future research priorities

The gaps identified in this review allow identification of research priorities that would substantially strengthen the evidence base in this field:

Longitudinal studies of cortisol and intestinal inflammation markers in dogs and cats during long-haul air transport, with serial measurements before, during and after the flight, and adequate control groups. The use of salivary cortisol or other non-invasive matrices would make these studies methodologically feasible without compromising the welfare of the studied animals.

Microbiota studies in cats under controlled environmental stress conditions, characterising both composition (16S rRNA) and metabolic function (functional metagenomics, faecal metabolomics). The current absence of data in this species is the most urgent gap to address given the volume of cats participating in international movements.

Controlled trials of psychobiotics and prebiotics in dogs and cats with more robust designs: larger sample sizes, placebo groups, measurement of behavioural, neuroendocrine and microbiota outcomes, and follow-up of baseline recovery after cessation of the intervention.

Metabolic risk studies in cats during and after transport, measuring hepatic triglycerides and hepatic enzymes in relation to hyporexia duration, body condition and journey length. These studies should stratify by body condition to determine whether the vulnerability threshold documented in experimental fasting is reproduced under real transport conditions.

Editorial self-audit 10-point verification

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